CONCAVITY OF THE ENTROPY OF DETERMINANTAL MEASURES ON TWO POINTS, AND THE SECOND VARIATION IN GENERAL

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Abstract. For a positive contraction Q on $\ell^2(E)$ with E finite, let P_Q be the determinantal probability measure on 2^E and $\mathrm{Ent}(Q)$ its Shannon entropy. Lyons conjectured in 2003 that Ent is concave in Q. We prove the conjecture when $|E| = 2$. For general E we write the second derivative of Ent along a line as a negative quadratic form in the derivatives of the principal minors of Q plus a sum, over subsets T with $|T| \geq 2$, of the second derivatives of the minors $\det Q_T$ weighted by Möbius sums of $\log P_Q$. For two-element T these weights are nonnegative by the Hadamard–Fischer inequality, which is what makes the case $|E| = 2$ work. At a reducible kernel $Q = Q_1 \oplus Q_2$ the second variation vanishes in every off-block direction, and the entropy decreases there at fourth order, by the χ^2 term of the mutual information between the two blocks. For blocks of sizes k and $n-k$ and $\|D\|_F = 1$ the fourth-order coefficient is at least $2/(k(n-k))$, with equality at $Q = \frac{1}{2}I$; the proof is a Cramér–Rao bound for a centred pair statistic combined with the nonnegativity of the inverse of the indicator covariance. We also report numerical evidence for $|E| \leq 5$, for real and complex kernels: the largest eigenvalue of the Hessian of Ent is negative at random kernels, and its supremum, zero, is approached at reducible kernels.

1. Introduction

Let E be a finite set and Q a positive contraction on $\ell^2(E)$, that is, a Hermitian matrix with $0 \leq Q \leq I$. The determinantal probability measure P_Q on 2^E is determined by $P_Q[\, A \subseteq Y \,] = \det Q_A$ for $A \subseteq E$, where Q_A is the principal submatrix indexed by A [1]. Its entropy is

$$\mathrm{Ent}(Q) \;=\; -\sum_{S \subseteq E} P_Q[Y = S] \log P_Q[Y = S].$$

In Section 9 of [1], Lyons asked whether Ent is concave on the convex set $\{0 \leq Q \leq I\}$ and stated the following conjecture, supported by numerical calculation.

Conjecture 1.1 (Lyons [1, Conjecture 9.2])**.** *For positive contractions Q_1, Q_2 on $\ell^2(E)$,* $\mathrm{Ent}\big((Q_1 + Q_2)/2\big) \geq \big(\mathrm{Ent}(Q_1) + \mathrm{Ent}(Q_2)\big)/2$.

The conjecture is restated as Conjecture 2.6 of Lyons's survey [2]. By finite restrictions it implies Conjecture 5.3 there: the entropy of the stationary determinantal process P_f with symbol f, studied by Lyons and Steif [3], is a concave function of f. The map $Q \mapsto P_Q$ is polynomial of degree $|E|$, so concavity of Ent is not inherited from the concavity of $x \mapsto -x \log x$.

Theorem 1.2. *Conjecture 1.1 holds when $|E| = 2$.*

Section 2 proves Theorem 1.2. Section 3 gives the second variation of Ent for arbitrary finite E (Proposition 3.1) and explains which part of the two-point argument survives. Section 4 shows that the second variation vanishes at reducible kernels in the off-block directions and identifies the fourth-order term, which Theorem 4.3 bounds below by $2\|D\|_F^4/(k(n-k))$, sharply. Section 5 records numerical evidence for $|E| = 3$ and 4.

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2. Two points

Let $E = \{1, 2\}$ and write $Q = \left(\begin{smallmatrix} a & b \\ \bar{b} & d \end{smallmatrix}\right)$. The four probabilities are

$$p_{12} = \det Q, \qquad p_1 = a - \det Q, \qquad p_2 = d - \det Q, \qquad p_0 = 1 - a - d + \det Q, \tag{2.1}$$

where $p_S = P_Q[Y = S]$ and $p_0 = P_Q[Y = \emptyset]$. A direct computation gives

$$p_1 p_2 - p_0 p_{12} = |b|^2. \tag{2.2}$$

Since Ent is continuous on the compact set $\{0 \le Q \le I\}$, it suffices to prove concavity on the open set $\{0 < Q < I\}$, where all four probabilities are positive: $p_0 = \det(I - Q) > 0$ and $p_{12} = \det Q > 0$, so $p_1 p_2 > 0$ by (2.2), and $p_1 + p_2 = \lambda_1 + \lambda_2 - 2\lambda_1\lambda_2 > 0$ for the eigenvalues $\lambda_1, \lambda_2 \in (0, 1)$ of Q. On that set concavity is equivalent to $f''(0) \le 0$ for every Q and every Hermitian D, where $f(t) = \mathrm{Ent}(Q + tD)$.

Put $\varphi(x) = -x \log x$ and

$$G(a, d, g) = \varphi(g) + \varphi(a - g) + \varphi(d - g) + \varphi(1 - a - d + g),$$

so that $\mathrm{Ent}(Q) = G(a, d, \det Q)$ by (2.1). The function G is a sum of concave functions of affine maps, hence concave, and

$$\frac{\partial G}{\partial g} = \log \frac{p_1 p_2}{p_0 p_{12}} =: L \ \ge 0 \tag{2.3}$$

by (2.2). Along $Q + tD$ the entries a, d move affinely, $\frac{d}{dt} \det(Q + tD)\big|_0 = \mathrm{tr}(\mathrm{adj}(Q) D) =: \gamma$ and $\frac{d^2}{dt^2} \det(Q + tD) = 2 \det D$. Writing $x = D_{11}$, $y = D_{22}$, the chain rule gives

$$f''(0) = -S + 2L \det D, \qquad S = \frac{\ell_{12}^2}{p_{12}} + \frac{\ell_1^2}{p_1} + \frac{\ell_2^2}{p_2} + \frac{\ell_0^2}{p_0}, \tag{2.4}$$

with $\ell_{12} = \gamma$, $\ell_1 = x - \gamma$, $\ell_2 = y - \gamma$, $\ell_0 = \gamma - x - y$. One checks directly that

$$\ell_1 \ell_2 - \ell_{12} \ell_0 = xy. \tag{2.5}$$

Lemma 2.1. *For $0 < Q < I$ on $\mathbb{C}^2$, the matrix $\mathrm{adj}\, Q$ has the same eigenvalues as Q; in particular $0 \le \mathrm{adj}\, Q \le I$.*

Proof. $\mathrm{adj}\, Q = (\det Q)\, Q^{-1}$ has eigenvalues $\lambda_1\lambda_2/\lambda_1 = \lambda_2$ and $\lambda_1\lambda_2/\lambda_2 = \lambda_1$. □

Lemma 2.2. *Let $0 < Q < I$ and $D \ge 0$ on $\mathbb{C}^2$, and put $A = \gamma\,(\mathrm{tr}\, D - \gamma) = -\ell_{12}\ell_0$ and $r = \sqrt{p_0 p_{12}}$. Then $A \ge (p_1 + p_2 + 2r) \det D$.*

Proof. Work in an orthonormal eigenbasis of $\mathrm{adj}\, Q$, with eigenvalues $\mu_1 = \lambda_2$, $\mu_2 = \lambda_1$ (Lemma 2.1), and let $\alpha, \delta \ge 0$ be the diagonal entries of D in that basis, so $\alpha\delta \ge \det D$. Then $\gamma = \mu_1\alpha + \mu_2\delta$ and $\mathrm{tr}\, D - \gamma = (1 - \mu_1)\alpha + (1 - \mu_2)\delta$, and

$$A = \mu_1(1 - \mu_1)\alpha^2 + \mu_2(1 - \mu_2)\delta^2 + \big[\mu_1(1 - \mu_2) + \mu_2(1 - \mu_1)\big]\alpha\delta.$$

The bracket equals $\lambda_1 + \lambda_2 - 2\lambda_1\lambda_2 = p_1 + p_2$, and by the arithmetic–geometric mean inequality the first two terms are at least $2\sqrt{\mu_1\mu_2(1 - \mu_1)(1 - \mu_2)}\,\alpha\delta = 2r\,\alpha\delta$, because $\lambda_1\lambda_2 = p_{12}$ and $(1 - \lambda_1)(1 - \lambda_2) = p_0$. Hence $A \ge (p_1 + p_2 + 2r)\alpha\delta \ge (p_1 + p_2 + 2r) \det D$. □

Proof of Theorem 1.2. Fix $0 < Q < I$ and a Hermitian D. If $\det D \le 0$ then $f''(0) \le 0$ by (2.4) and (2.3). Otherwise D is definite, and since (2.4) is even in D we may assume $D \ge 0$. Put $s = \sqrt{p_1 p_2}$ and $r = \sqrt{p_0 p_{12}}$, so $s \ge r$ by (2.2). Applying $u^2/p + v^2/q \ge 2|uv|/\sqrt{pq}$ to the pairs (ℓ_{12}, ℓ_0) and (ℓ_1, ℓ_2),

$$S \ \ge \ \frac{2|\ell_{12}\ell_0|}{r} + \frac{2|\ell_1\ell_2|}{s} = \frac{2(|A|s + |B|r)}{rs}, \qquad B := \ell_1\ell_2.$$

The logarithmic mean of s^2 and r^2 is at least their geometric mean, so $L = \log(s^2/r^2) \le (s^2 - r^2)/(rs)$. By (2.4) it therefore suffices to show

$$|A|s + |B|r \;\ge\; (s^2 - r^2)\det D. \tag{2.6}$$

By Lemma 2.1 and $D \ge 0$ we have $\gamma \ge 0$ and $\operatorname{tr} D - \gamma = \operatorname{tr}((I - \operatorname{adj} Q)D) \ge 0$, so $A \ge 0$; and $A + B = xy \ge \det D \ge 0$ by (2.5).

If $B < 0$, then $A > xy \ge \det D$, and $s \le \frac{1}{2}(p_1 + p_2) \le 1$ gives $As \ge s^2 \det D \ge (s^2 - r^2)\det D$.

If $B \ge 0$, then $As + Br = (s - r)A + r\,xy$. Lemma 2.2 and $p_1 + p_2 \ge 2s$ give $A \ge (2s + 2r)\det D \ge (s + r)\det D$, hence $As + Br \ge (s - r)(s + r)\det D + r\,xy \ge (s^2 - r^2)\det D$.

In both cases (2.6) holds, so $f''(0) \le 0$. Thus Ent is concave on $\{0 < Q < I\}$ and, by continuity, on $\{0 \le Q \le I\}$. □

Remark 2.3. The bound is far from tight in the middle of the parameter range. Over 2×10^5 random pairs (Q, D) with D definite, the ratio $2L \det D/S$ never exceeded 0.28. Equality $f''(0) = 0$ occurs, for instance, at $b = 0$, $a = d$ (that is, $Q = cI$) with D off-diagonal: there P_Q is a product measure, and moving Q off the diagonal keeps the one-point marginals fixed while P_Q changes only at second order, so the entropy, which is maximal among laws with those marginals, changes at fourth order.

3. The second variation for general E

For $T \subseteq E$ write $m_T = \det Q_T$ (with $m_\emptyset = 1$). Inclusion–exclusion gives

$$P_Q[Y = S] = \sum_{T \supseteq S} (-1)^{|T \setminus S|} m_T, \tag{3.1}$$

so $P_Q[Y = \cdot\,]$ is an affine function of the vector of principal minors. Differentiating $\mathrm{Ent} = \sum_S \varphi(P_Q[Y = S])$ twice along $Q + tD$ gives the following.

Proposition 3.1. *Let $0 < Q < I$, D Hermitian, $p_S = P_Q[Y = S]$, and m'_T, m''_T the first two derivatives of $\det(Q + tD)_T$ at $t = 0$. Then*

$$\frac{d^2}{dt^2}\,\mathrm{Ent}(Q + tD)\Big|_{t=0} = -\sum_{S \subseteq E} \frac{1}{p_S}\Big(\sum_{T \supseteq S} (-1)^{|T \setminus S|} m'_T\Big)^2 + \sum_{|T| \ge 2} \Lambda_T\, m''_T,$$

where $\Lambda_T = -\sum_{S \subseteq T} (-1)^{|T \setminus S|} \log p_S$.

Proof. By (3.1), $\frac{d}{dt} p_S = \sum_{T \supseteq S} (-1)^{|T \setminus S|} m'_T$ and $\frac{d^2}{dt^2} p_S = \sum_{T \supseteq S} (-1)^{|T \setminus S|} m''_T$. Since $\varphi'(x) = -1 - \log x$ and $\varphi''(x) = -1/x$,

$$\frac{d^2}{dt^2}\,\mathrm{Ent} = \sum_S \varphi''(p_S)\Big(\frac{dp_S}{dt}\Big)^2 + \sum_T m''_T \sum_{S \subseteq T} (-1)^{|T \setminus S|} \varphi'(p_S).$$

The constant -1 in φ' contributes $\sum_{S \subseteq T} (-1)^{|T \setminus S|} = 0$ for $T \ne \emptyset$, and $m''_T = 0$ for $|T| \le 1$ because m_T is then affine in t. □

When $Q < I$, the measure P_Q is the L-ensemble with $L = Q(I - Q)^{-1}$, that is $p_S = \det L_S / \det(I + L)$. The normalisation cancels in Λ_T for $T \ne \emptyset$, so

$$\Lambda_T = -\sum_{S \subseteq T} (-1)^{|T \setminus S|} \log\det L_S.$$

For $T = \{i, j\}$,

$$\Lambda_{ij} = \log \frac{L_{ii} L_{jj}}{\det L_{\{i,j\}}} \;\ge 0$$

by the Hadamard–Fischer inequality; for $|E| = 2$ this is (2.3). The proof of Theorem 1.2 uses exactly two features of the case $|E| = 2$: the only curvature term is $\Lambda_{12}\, m''_{12} = 2\Lambda_{12} \det D$, with $\Lambda_{12} \geq 0$, and Λ_{12} is controlled by the quadratic form through the logarithmic-mean inequality. For $|E| \geq 3$ the coefficients Λ_T with $|T| \geq 3$ take both signs: for $|E| = 3$, in 3000 random kernels with spectra in $[0.05, 0.95]$, Λ_{123} was negative in 1508 cases, and $\Lambda_T m''_T > 0$ occurred for both $|T| = 2$ and $|T| = 3$. The curvature terms therefore have to be compared with the quadratic form jointly.

4. Reducible kernels

The equality case of Remark 2.3 is an instance of a general phenomenon. Call Q *reducible* if $E = B_1 \sqcup B_2$ with B_1, B_2 nonempty and $Q = Q_1 \oplus Q_2$, where $Q_j = Q_{B_j}$. Then $P_Q = P_{Q_1} \otimes P_{Q_2}$, since $\det Q_A = \det Q_{A\cap B_1} \det Q_{A\cap B_2}$ for every $A \subseteq E$. An *off-block direction* is a Hermitian D with $D_{B_1} = 0$ and $D_{B_2} = 0$.

Lemma 4.1. *Let $Q = Q_1 \oplus Q_2$ and let D be an off-block direction. For every real t with $0 \leq Q + tD \leq I$, the law of $Y \cap B_j$ under P_{Q+tD} is P_{Q_j}, $j = 1, 2$. Consequently*

$$\mathrm{Ent}(Q + tD) = \mathrm{Ent}(Q_1) + \mathrm{Ent}(Q_2) - I(t),$$

where $I(t)$ is the mutual information of $Y \cap B_1$ and $Y \cap B_2$ under P_{Q+tD}, and $I(t)$ is the relative entropy of P_{Q+tD} with respect to P_Q.

Proof. For $A \subseteq B_j$ we have $P_{Q+tD}[A \subseteq Y] = \det(Q + tD)_A = \det(Q_j)_A$, because D vanishes on $B_j \times B_j$; these probabilities determine the law of $Y \cap B_j$. The identity for Ent is the chain rule $H(Y) = H(Y \cap B_1) + H(Y \cap B_2) - I$. Since both marginals are fixed and P_Q is their product, the mutual information equals the relative entropy of P_{Q+tD} with respect to P_Q. □

Proposition 4.2. *Let $0 < Q < I$ be reducible as above, D an off-block direction, $p_S = P_Q[Y = S]$ and $p_S(t) = P_{Q+tD}[Y = S]$. Then $p_S(t) = p_S + r_S t^2 + O(t^4)$ for some real r_S, and*

$$\mathrm{Ent}(Q + tD) = \mathrm{Ent}(Q) - \frac{t^4}{2} \sum_{S\subseteq E} \frac{r_S^2}{p_S} + O(t^6).$$

In particular the second derivative of Ent *at Q in the direction D is 0, and $t \mapsto \mathrm{Ent}(Q + tD)$ has a strict local maximum at $t = 0$ unless $r = 0$.*

Proof. Let $U = I_{B_1} \oplus (-I_{B_2})$. Then $U(Q + tD)U^* = Q - tD$, and conjugation by a diagonal unitary does not change principal minors, so $p_S(-t) = p_S(t)$; each $p_S(t)$ is a polynomial in t, hence $p_S(t) = p_S + r_S t^2 + O(t^4)$. Since $0 < Q < I$, P_Q is the L-ensemble with $L = Q(I - Q)^{-1}$ positive definite, so every $p_S > 0$. By Lemma 4.1, $\mathrm{Ent}(Q) - \mathrm{Ent}(Q+tD) = I(t) = \sum_S p_S(t) \log\big(p_S(t)/p_S\big)$. With $\delta_S = p_S(t) - p_S$ and $\sum_S \delta_S = 0$,

$$\sum_S (p_S + \delta_S) \log\Big(1 + \frac{\delta_S}{p_S}\Big) = \sum_S \frac{\delta_S^2}{2p_S} + O(|\delta|^3),$$

and $\delta_S = r_S t^2 + O(t^4)$. □

For $|E| = 2$ and $Q = \mathrm{diag}(q_1, q_2)$, $D = \left(\begin{smallmatrix} 0 & x \\ \bar{x} & 0 \end{smallmatrix}\right)$, the four atoms are $p_S \pm t^2|x|^2$ and the coefficient is $\frac{1}{2}|x|^4 \sum_S p_S^{-1}$; for $\|D\|_F = 1$ and $q_1 = q_2 = \frac{1}{2}$ it equals 2. Two consequences are worth recording. First, the Hessian of Ent on $\{0 < Q < I\}$ is singular at every reducible kernel, so a proof of Conjecture 1.1 cannot rest on a uniform negative bound for the second variation; near reducible kernels the fourth-order term is the one that carries the sign. Second, the lemma gives more than the local statement: along every off-block line through a reducible kernel, Ent is maximal

at the reducible kernel itself, because $I(t) \geq 0$. Concavity along such a line, however, would need I to be convex in t, which Lemma 4.1 does not give.

The fourth-order coefficient is bounded below uniformly in Q and $|E|$.

Theorem 4.3. *In the setting of Proposition 4.2, let $|B_1| = k$, $|B_2| = n - k$ and $x_{ij} = |D_{ij}|^2$ for $i \in B_1$, $j \in B_2$. Then*

$$\frac{1}{2}\sum_{S \subseteq E} \frac{r_S^2}{p_S} \;\geq\; 8 \sum_{i \in B_1,\, j \in B_2} x_{ij}^2 \;\geq\; \frac{2\,\|D\|_F^4}{k(n-k)}.$$

Both inequalities are equalities when $Q = \frac{1}{2}I$ and $|D_{ij}|$ is constant on $B_1 \times B_2$.

Proof. Write $\xi_i = \mathbf{1}[i \in Y]$, $q_i = Q_{ii}$, and let $C^{(j)}$ be the covariance matrix of $(\xi_i)_{i \in B_j}$ under P_{Q_j}, so that $C^{(j)}_{ii} = q_i(1 - q_i) \leq \frac{1}{4}$ and $C^{(j)}_{ii'} = -|Q_{ii'}|^2 \leq 0$ for $i \neq i'$. A linear combination of the ξ_i, $i \in B_j$, that is constant on 2^{B_j} is zero (compare $\emptyset$ with the singletons), and P_{Q_j} charges every subset of B_j, so $C^{(j)}$ is positive definite.

Step 1. Since $\sum_S r_S = 0$ and $p_S > 0$, for every function f on 2^E the Cauchy–Schwarz inequality gives

$$\Big(\sum_S r_S f(S)\Big)^2 = \Big(\sum_S r_S (f(S) - E_{P_Q} f)\Big)^2 \leq \Big(\sum_S \frac{r_S^2}{p_S}\Big)\,\mathrm{Var}_{P_Q}(f). \tag{4.1}$$

Step 2. For a real $B_1 \times B_2$ matrix W put $f = \sum_{i \in B_1, j \in B_2} W_{ij}(\xi_i - q_i)(\xi_j - q_j)$. Under P_{Q+tD} we have $P[\xi_i = 1] = q_i$, because D has zero diagonal, and for $i \in B_1$, $j \in B_2$,

$$P_{Q+tD}[\xi_i = \xi_j = 1] = \det \begin{pmatrix} q_i & tD_{ij} \\ t\bar{D}_{ij} & q_j \end{pmatrix} = q_i q_j - t^2 x_{ij}.$$

Hence $E_{P_{Q+tD}} f = -t^2 \sum W_{ij} x_{ij}$ and $\sum_S r_S f(S) = -\sum W_{ij} x_{ij}$. Under $P_Q = P_{Q_1} \otimes P_{Q_2}$ the two blocks are independent, so $\mathrm{Var}_{P_Q}(f) = \mathrm{tr}\big(C^{(1)} W C^{(2)} W^{\mathsf{T}}\big)$.

Step 3. The matrix $C^{(1)}$ has nonpositive off-diagonal entries and is positive definite, so its inverse has nonnegative entries: with $s = \max_i C^{(1)}_{ii}$ and $N = I - C^{(1)}/s$, the matrix N is entrywise nonnegative and symmetric, its spectral radius is its largest eigenvalue $1 - \lambda_{\min}(C^{(1)})/s < 1$, and $(C^{(1)})^{-1} = s^{-1} \sum_{m \geq 0} N^m$. Let $V = (x_{ij})$ and take $W = (C^{(1)})^{-1} V$. Then $\sum W_{ij} x_{ij} = \mathrm{tr}\, G$ and $\mathrm{Var}_{P_Q}(f) = \mathrm{tr}(G C^{(2)})$, where $G = V^{\mathsf{T}} (C^{(1)})^{-1} V$ is positive semidefinite with nonnegative entries. As the off-diagonal entries of $C^{(2)}$ are nonpositive, $\mathrm{tr}(G C^{(2)}) \leq \sum_j G_{jj} C^{(2)}_{jj} \leq \frac{1}{4}\,\mathrm{tr}\, G$, and (4.1) gives $\sum_S r_S^2/p_S \geq 4\,\mathrm{tr}\, G$.

Step 4. Let v_j be the j-th column of V. By Cauchy–Schwarz in the inner product defined by $C^{(1)}$, and again because $v_j \geq 0$ entrywise and the off-diagonal entries of $C^{(1)}$ are nonpositive,

$$v_j^{\mathsf{T}} (C^{(1)})^{-1} v_j \geq \frac{|v_j|^4}{v_j^{\mathsf{T}} C^{(1)} v_j} \geq \frac{|v_j|^4}{\sum_i C^{(1)}_{ii} x_{ij}^2} \geq 4 |v_j|^2.$$

Summing over j, $\mathrm{tr}\, G \geq 4 \sum x_{ij}^2$, which is the first inequality. The second follows from $\|D\|_F^2 = 2 \sum x_{ij}$ and the Cauchy–Schwarz inequality over the $k(n-k)$ pairs. When $Q = \frac{1}{2}I$, $C^{(1)}$ and $C^{(2)}$ are $\frac{1}{4}I$ and all inequalities above become equalities for constant x_{ij}, except possibly (4.1). There, with M_S and J_S as in Section 5, $p_S(t) = \det(\frac{1}{2}I + t J_S D) = 2^{-n} \det(I + 2t J_S D)$; since $\mathrm{tr}(J_S D) = 0$ and $\mathrm{tr}(J_S D J_S D) = 2 \sum_{i \in B_1, j \in B_2} (J_S)_{ii} (J_S)_{jj} x_{ij}$, this gives $r_S/p_S = -4 \sum (J_S)_{ii} (J_S)_{jj} x_{ij} = -16 f(S)$ for $W = V$, because $(J_S)_{ii} = 2(\xi_i - \frac{1}{2})$. So r_S/p_S is proportional to $f - E_{P_Q} f$ and (4.1) is an equality too. $\square$

For $n = 2$ the bound is the value 2 computed above. Theorem 4.3 says that along any off-block line through a reducible kernel with $\|D\|_F = 1$, $\mathrm{Ent}(Q + tD) \le \mathrm{Ent}(Q) - 2t^4/(k(n-k)) + O(t^6)$; the constant does not degenerate as Q approaches the boundary of $\{0 < Q < I\}$ or as the blocks become ill-conditioned.

5. Numerical evidence for three and four points

We searched for counterexamples in two ways, with kernels $Q = U\,\mathrm{diag}(\lambda)U^*$ for U from the QR factorisation of a Gaussian matrix (real or complex) and $\lambda \in (0,1)^n$.

Midpoint gaps. For $n = |E| = 3$ (real) and $n = 4$ (complex) we minimised $\mathrm{Ent}\big((Q_1 + Q_2)/2\big) - \big(\mathrm{Ent}(Q_1) + \mathrm{Ent}(Q_2)\big)/2$ by Nelder–Mead from the worst of 2000 and 3000 random pairs. The minima were 8×10^{-15} and 2×10^{-8}, attained as $Q_1 - Q_2 \to 0$.

Second derivatives. We maximised the central difference $\big(\mathrm{Ent}(Q + hD) + \mathrm{Ent}(Q - hD) - 2\,\mathrm{Ent}(Q)\big)/h^2$ with $h = 10^{-3}$ and $\|D\|_F = 1$, keeping the spectrum of Q at distance at least $5h$ from $\{0, 1\}$. For $n = 3$ (real) and $n = 4$ (complex) the maxima were -2×10^{-6} and -1.2×10^{-6}, approached only near scalar kernels $Q = cI$ (in every optimised kernel the eigenvalues agreed to within 4×10^{-4}). Near those points, for $n = 3$, $Q = cI + \varepsilon A$ with $c \in [0.2, 0.8]$, ε down to 3×10^{-3} and 400 random real symmetric pairs (A, D), half of them with D off-diagonal, the second derivative stayed negative. This matches Remark 2.3: scalar kernels give product measures, and off-diagonal perturbations preserve the one-point marginals.

The formula of Proposition 3.1 agrees with central differences to relative accuracy 3×10^{-6} at $n = 3$.

The exact Hessian. For $S \subseteq E$ let M_S be the matrix whose i-th row is the i-th row of Q if $i \in S$ and of $I - Q$ if $i \notin S$; by multilinearity of the determinant in rows, $p_S = \det M_S$. Along $Q + tD$ we have $M_S' = J_S D$ with J_S diagonal, $+1$ on S and -1 off S. For a basis (E_a) of the Hermitian matrices and $A_a = M_S^{-1} J_S E_a$, Jacobi's formula gives $\partial_a p_S = p_S \operatorname{tr} A_a$ and $\partial_a \partial_b p_S = p_S(\operatorname{tr} A_a \operatorname{tr} A_b - \operatorname{tr} A_a A_b)$, so the Hessian of Ent is

$$H_{ab} = -\sum_S \Big(\partial_a \partial_b p_S \, \log p_S + \frac{\partial_a p_S \, \partial_b p_S}{p_S} \Big).$$

This agrees with finite differences to relative accuracy 10^{-6} for real and 10^{-7} for complex kernels, $n = 2, 3, 4$. Writing $Q = U\,\mathrm{diag}(\sigma(x))U^*$ with σ the logistic function and U the exponential of a real skew-symmetric (respectively skew-Hermitian) matrix, we maximised the largest eigenvalue of H over (x, U) by Nelder–Mead. Over real symmetric directions, with 8, 20, 10 and 3 starts for $n = 2, 3, 4, 5$, the largest values found were 3.8×10^{-15}, 3.6×10^{-15}, 7.1×10^{-15} and 3.3×10^{-15}; over Hermitian directions, with 6, 8, 6 and 3 starts, 1.6×10^{-15}, 3.2×10^{-15}, 4.1×10^{-15} and 2.4×10^{-15}. These are zero to rounding, and every maximiser was reducible ($n = 2$: diagonal; $n = 3, 4, 5$: one point split off from the rest, with off-block entries below 10^{-7} at $n = 5$), which is the null direction of Proposition 4.2. At random kernels, without optimisation, the largest eigenvalue was strictly negative (median -0.10 for $n = 3$ and -0.034 for $n = 4$ over 40 real kernels).

Fourth order at reducible kernels. For 48 reducible kernels with random off-block directions $\|D\|_F = 1$ ($n = 2, \dots, 5$, real and complex, random block sizes), a least-squares fit of $\mathrm{Ent}(Q + tD) - \mathrm{Ent}(Q) = at^2 + bt^4 + ct^6$ on $t \in [0.02, 0.06]$ gave $|a| \le 10^{-6}$ and $b < 0$ in every case, the largest being $b = -0.536$, in agreement with Proposition 4.2. The quantity $\frac{1}{2}\sum_S r_S^2/p_S$, computed from the t^2 coefficients of the atoms, has infimum 2, 1 and 0.5 for $n = 2$, for $n = 3$ with blocks of sizes $1, 2$, and for $n = 4$ with two blocks of size 2, approached as the spectrum of Q tends to $\{\frac{1}{2}\}$; these are the values $2/(k(n-k))$ of Theorem 4.3. In 60 random reducible kernels for each of $(n, k) = (3, 1), (4, 2), (5, 2), (6, 3)$, real and complex, the ratio of $\sum_S r_S^2/p_S$

to the Cramér–Rao bound of Step 2 with the optimal W was at least 1, and that bound was at least 1.004 times $16\sum x_{ij}^2$. Minimising the ratio of $\sum_S r_S^2/p_S$ to $16\sum x_{ij}^2$ over reducible Q and off-block D by Nelder–Mead, with 6 starts for $(n,k) = (3,1)$ and 4 for $(4,1)$ and $(4,2)$, real kernels, gave 1.000000 in every case: the search finds the equality case of Theorem 4.3 and nothing below it.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [4] and working together inside the Brain framework [6], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [5], the Math Brain repository [4] and the Brain framework repository [6].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. In November 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, found the proof of Theorem 1.2, the formula of Proposition 3.1, Proposition 4.2 and Theorem 4.3, wrote and ran the computations of Section 5, and wrote the paper. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code for the computations of Remark 2.3 and Sections 3–5, with their outputs, is available at
`https://github.com/yspennstate/lyons-determinantal-entropy-two-points`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [4].

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